JURNAL PRAKTIKUM ELEKTRONIKA ANALOG

Op-Amp Rangkaian Inverting dan Non-Inverting

Tanggal Praktikum : Rabu, 26 Oktober 2022

Kelompok : 3 Gabungan : Kel. 4

Nama Anggota : 1. Anisha Fitri Pertiwi (211344005) 3 Bagoa Alifa (211344007)

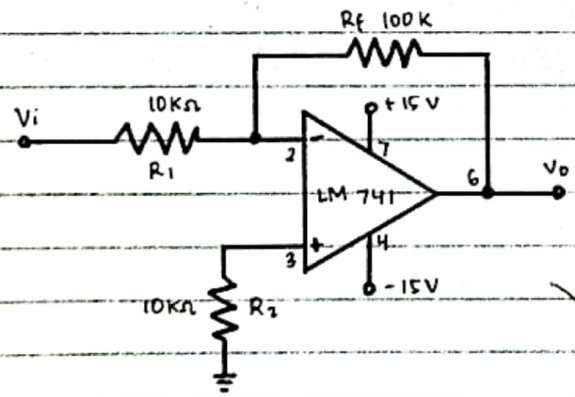
2. Aulia Nadeif Pramesti (211344006)

Kelas : 2NK

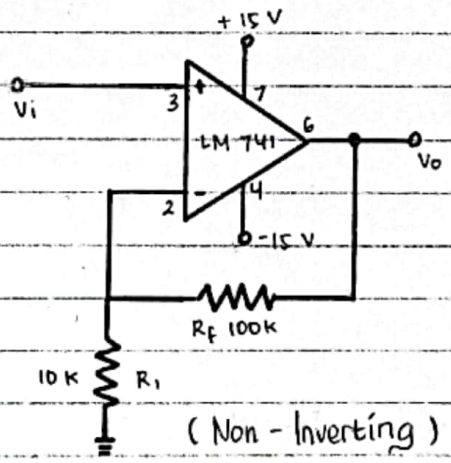
ALAT DAN BAHAN

1. DC Power Supply (2)
2. Function Generator (1)
3. Oscilloscope (1)
4. Op-Amp 741 (1)
5. $R_1 = R_2 = 10K\Omega$; $R_f = 100K\Omega$
6. Multimeter (1)
7. Kabel Jumper (secukupnya)
8. Konektor Jumper (secukupnya)
9. Kertas HVS dan millimeter block (1 lembar / kelompok)

RANGKAIAN SKEMATIK



(Inverting)



(Non - Inverting)

PERHITUNGAN TEORI

1) Inverting

$$Vo = -\frac{Rf}{R_1} \cdot Vi = -\frac{100\,K\Omega}{10\,K\Omega} \cdot 10\,mV = -100\,mV$$

$$Av = -\frac{Rf}{R_1} = -\frac{100\,K\Omega}{10\,K\Omega} = -10X$$

Non - Inverting

$$Vo = \left(\frac{Rf}{Ri} + 1\right) \cdot Vi = \left(\frac{100\,K\Omega}{10\,K\Omega} + 1\right) \cdot 10mV = 110\ mV$$

$$Av = \frac{Rf}{Ri} + 1 = \frac{100\,K\Omega}{10\,K\Omega} + 1 = 11\ x$$

TABEL PENGUKURAN

1) Inverting

Vi (mV)	Vo (mV)			Av (x)		
	Teori	Praktek	% Error	Teori	Praktek	% Error
10 mV	-100	-108	8 %	-10	-10,8	8 %

Vo Praktek
20 mV/div x 5,4 kotak Vertikal
= -108 mV
↓
Beda Phasa : 180°

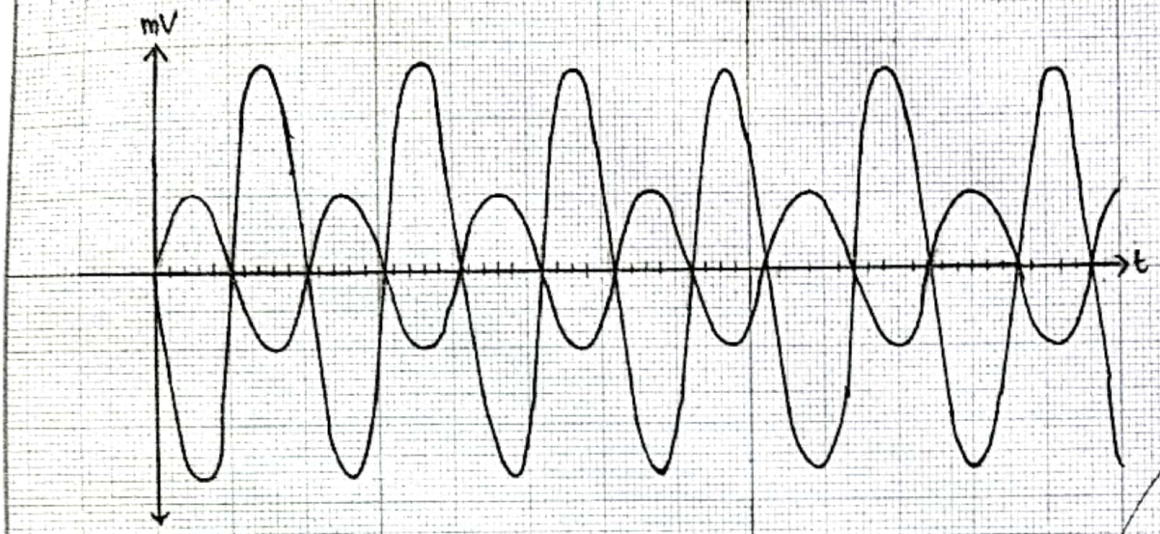
2) Non - Inverting

Vi (mV)	Vo (mV)			Av (x)		
	Teori	Praktek	% Error	Teori	Praktek	% Error
10MV	110	120	9 %	11	12	9 %

Vo Praktek
20 mV/div x 6 kotak Vertikal
= 120 mV

GAMBAR SINYAL INPUT DAN OUTPUT

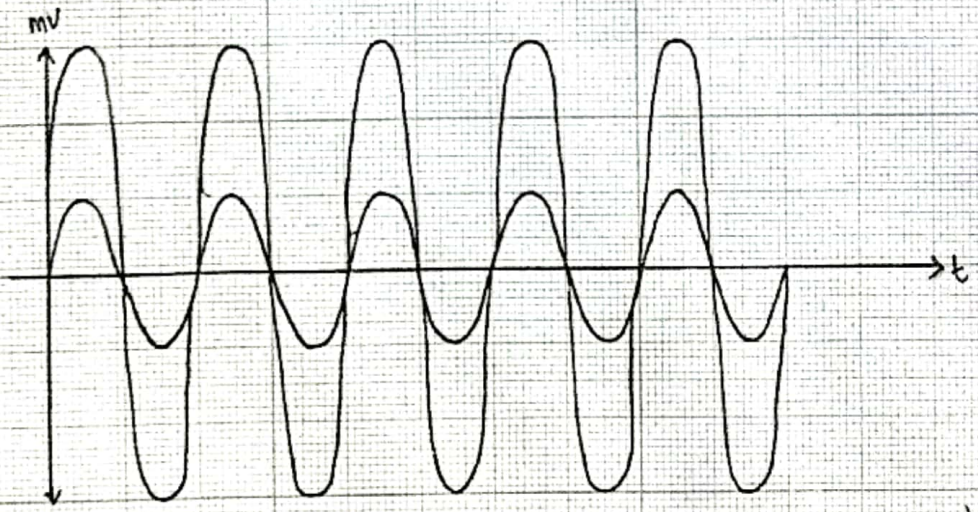
A. GAMBAR SINYAL PADA RANGKAIAN INVERTING



Volt /div :
V_o = 20mV
V_i = 5mV

Time /div :
0,5 ms

B. GAMBAR SINYAL PADA RANGKAIAN NON-INVERTING



Volt /div :
V_o = 20mV
V_i = 5mV

Time /div :
0,5 ms

JURNAL PRAKTIKUM ELEKTRONIKA ANALOG

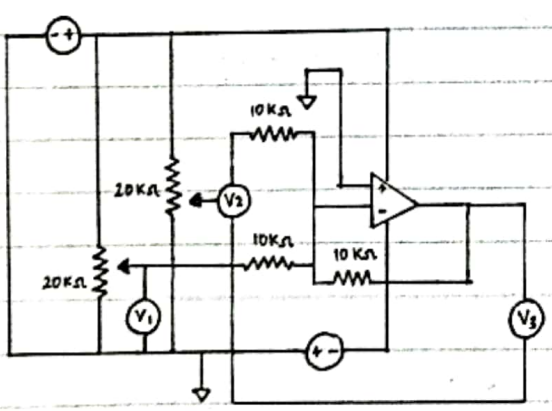
Aplikasi Op-Amp Rangkaian Summing dan Differential

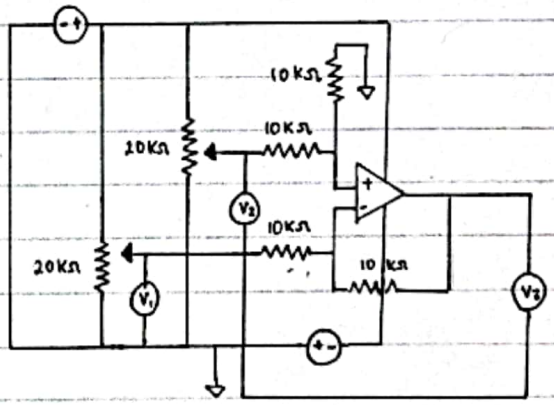
Tanggal Praktikum : Rabu, 9 November 2022
Kelompok : 3
Nama Anggota : 1. Anisha Fitri Pertiwi (2113440005)
2. Aulia Nadzif Pramesti (2113440006)
Kelas : 2NK

ALAT DAN BAHAN

1. DC Power Supply (2)
2. Oscilloscope (1)
3. Op-Amp 741 (1)
4. R = 10KΩ (5) ; R=22KΩ (1) ; R = 8K2Ω (1) ; R = 2K2Ω (1) ; R = 1KΩ (1) ; R potensio R = 20 / 50 KΩ (2)
5. Multimeter (1)
6. Kabel Jumper (secukupnya)
7. Konektor Jumper (secukupnya)
8. Kertas HVS dan milimeter block (masing-masing 1 lembar/kelompok)

RANGKAIAN SKEMATIK





SUMMING

R_1 = 10K
R_2 = 10K
R_f = 10K
R_{po1} = 20K
R_{po2} = 10K

V_{CC+} = +10V
V_{CC-} = −10V

DIFFERENTIAL

R_1 = 10K
R_2 = 10K
R_3 = 10K
R_f = 10K
R_{po1} = 20K
R_{po2} = 20K

V_{CC+} = +10V
V_{CC-} = −10V

Paraf

PERHITUNGAN

• Untuk (V_1(V)) = 1V

$$V_1 = \frac{R}{R+10K} \cdot VCC$$

$$1V = \frac{R}{R+10K} \cdot 10V$$

$$\frac{1V}{10V} = \frac{R}{R+10K}$$

$$\frac{R+10K}{10R} = 0$$

$$\frac{R+10K}{10} = R$$

$$0,1R + 1K = R$$

$$1K = R - 0,1R$$

$$R = \frac{1K}{0,9}$$

$$R = 1,1K$$

• Untuk (V_1(V)) = 2V

$$V_1 = \frac{R}{R+10K} \cdot VCC$$

$$2V = \frac{R}{R+10K} \cdot 10V$$

$$\frac{2V}{10V} = \frac{R}{R+10K}$$

$$\frac{2R+20K}{10} = R$$

$$0,2R + 2K = R$$

$$2K = R - 0,2R$$

$$R = \frac{2K}{0,8}$$

$$R = 2,5K$$

• Untuk (V_2(V)) = 2V

$$V_2 = \frac{R}{R+10K} \cdot 10V$$

$$\frac{2V}{10V} = \frac{R}{R+10K}$$

$$\frac{2R+20K}{10} = R$$

$$0,2R + 2K = R$$

$$2K = R - 0,2R$$

$$R = \frac{2K}{0,8}$$

$$R = 2,5K$$

• Untuk (V_2(V)) = 3V

$$V_2 = \frac{R}{R+10K} \cdot 10V$$

$$\frac{3V}{10V} = \frac{R}{R+10K}$$

$$\frac{3R+30K}{10} = R$$

$$0,3R + 3K = R$$

$$3K = R - 0,3R$$

$$R = \frac{3K}{0,7}$$

$$R = 4,2K$$

untuk V_2 (V) = 5

$$V_2 = \frac{R}{R+10K} \cdot VCC$$

$$\frac{5}{10} = \frac{R}{R+10K}$$

$$\frac{5R+50K}{10} = R$$

$$0{,}5R + 5K = R$$

$$5K = R - 0{,}5R$$

$$5K = 0{,}5R$$

$$R = \frac{5K}{0{,}5}$$

$$R = 10K$$

Bila $R_1 = R_2 = R_f$

Vout = - (V1 + V2) → Summing

Vout = V2 - V1 → Differential

PERCOBAAN 1 → SUMMING

1) Vout = - (1+2) = -3

2) Vout = - (1+3) = -4

3) Vout = - (2+5) = -7

PERCOBAAN 2 → DIFFERENTIAL

1) Vout = 2 - 1 = 1

2) Vout = 3 - 1 = 2

3) Vout = 5 - 2 = 3

TABEL PERCOBAAN

Summing	V1	V2	V3	Rpo 1	Rpo 2
Teori	1	2	-3	1,1K	2,5K
Praktek	1	2	-3	1,3K	2,8K
% Error	0	0	0	18	21,47
Teori	1	3	-4	1,1K	4,2K
Praktek	1	3	-4	1,3K	3,9K
% Error	0	0	0	18	7,69
Teori	2	5	-7	2,5K	10K
Praktek	2	5	-7,2	2,3K	2,2K
% Error	0	0	2,1	8,69	78

volt/div ↑ -1,5 × 2 = -3 →

volt/div ↑ -2 × 2 = -4 →

volt/div ↑ -3,6 × 2 = -7,2 →

Differential	V1	V2	V3	Rpo 1	Rpo 2
Teori	1	2	1	1,1K	2,5K
Praktek	1	2	1,2	1,3K	2,6K
% Error	0	0	16,6	18	3,84
Teori	1	3	2	1,1K	4,2K
Praktek	1	3	2	1,3K	3,9K
% Error	0	0	0	18	7,69
Teori	2	5	3	2,5K	10K
Praktek	2	5	3,2	2,3K	2,2K
% Error	0	0	6,6	8,69	78

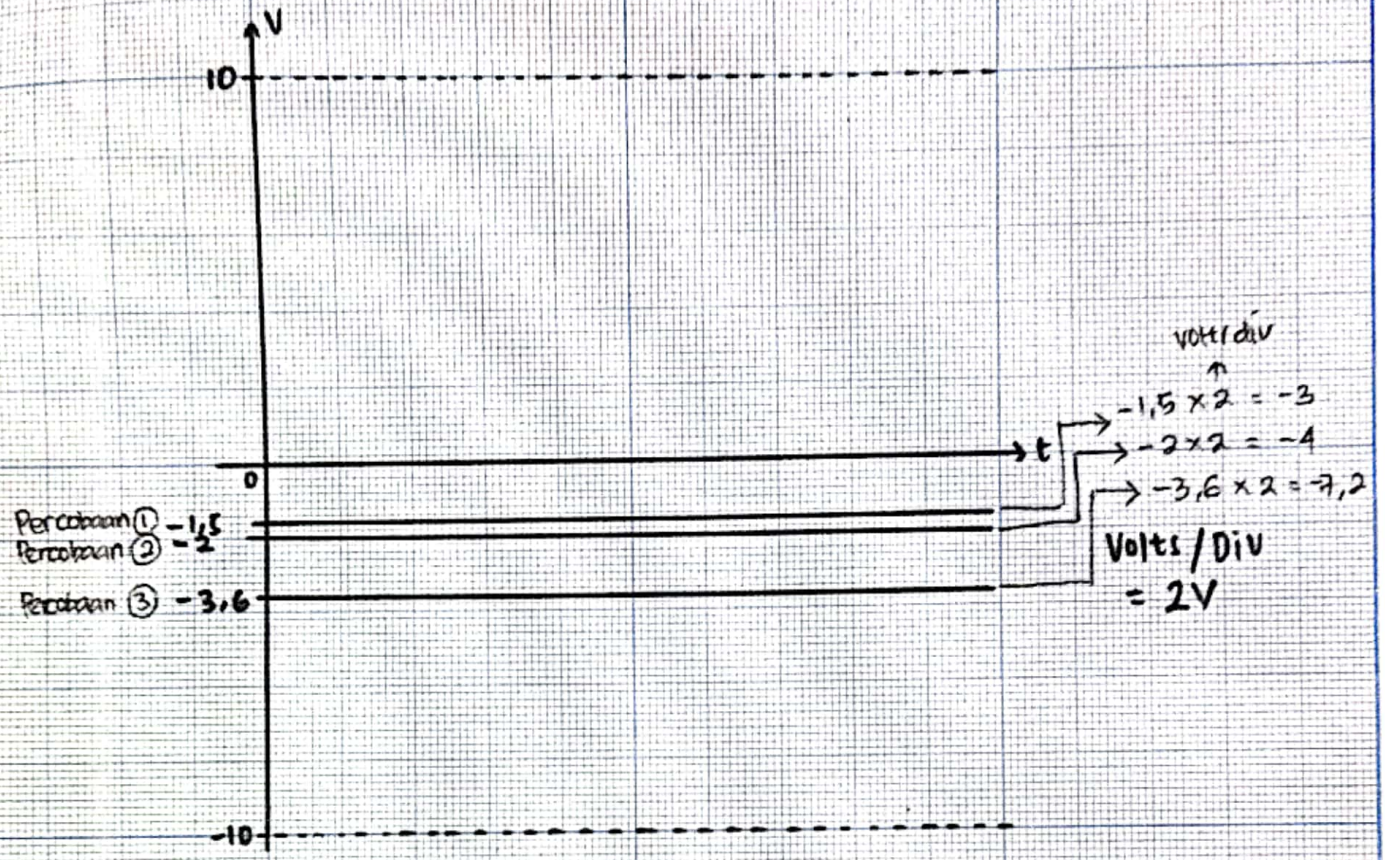
volt/div ↑ → 0,6 × 2 = 1,

volt/div ↑ → 1 × 2 = 2

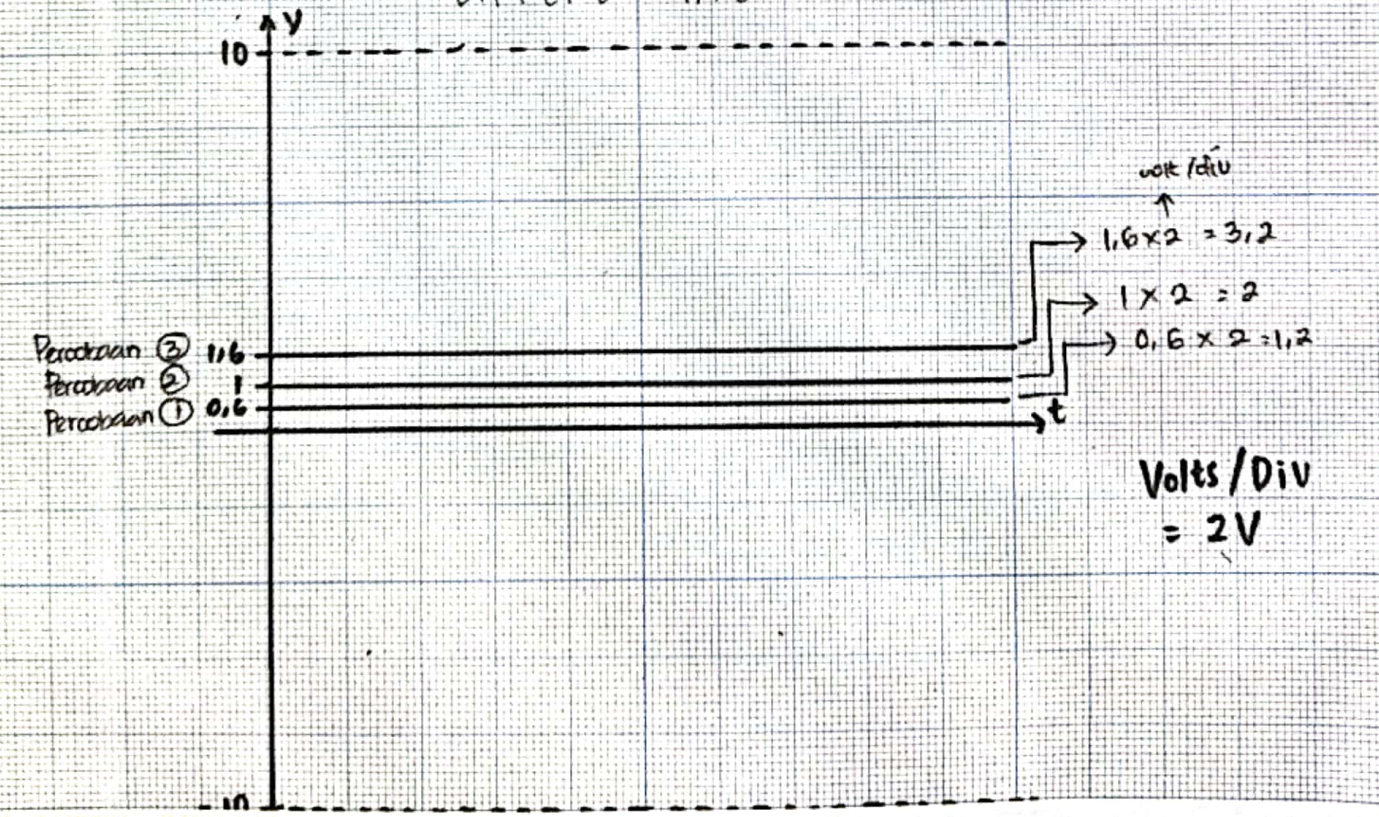
volt/div ↑ → 1,6 × 2 = 3,

SINYAL OUTPUT PERCOBAAN SUMMING

1. Amisha F P (21134400S)
2. Aulia N P (21134400G) } 2NK



SINYAL OUTPUT PERCOBAAN DIFFERENTIAL



JURNAL PRAKTIKUM ELEKTRONIKA ANALOG

Aplikasi Op-Amp Rangkaian Komparator

Acc 16/11/22

Tanggal Praktikum : Rabu, 16 November 2022

Kelompok : 3

Nama Anggota : 1. Anisha Fitri Pertiwi (21134400 5)
2. Aulia Nadzif Pramesti (211344006)
3. Bagea Alifa
4. Harun Maulana
5. Rezan Candrawinata

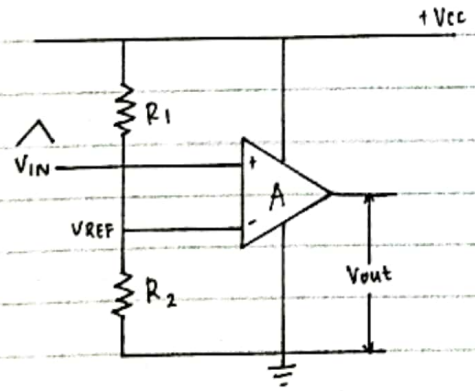
Kelas : 2NK

ALAT DAN BAHAN

1. DC Power Supply (2)
2. Function Generator (1)
3. Oscilloscope (1)
4. Op-Amp 741 (1)
5. R = 10KΩ (1) ; R = 1KΩ ;
6. Multimeter (1)
7. Kabel Jumper (secukupnya)
8. Konektor Jumper (secukupnya)
9. Kertas HVS dan milimeter block (masing-masing 1 lembar / kelompok)

RANGKAIAN SKEMATIK

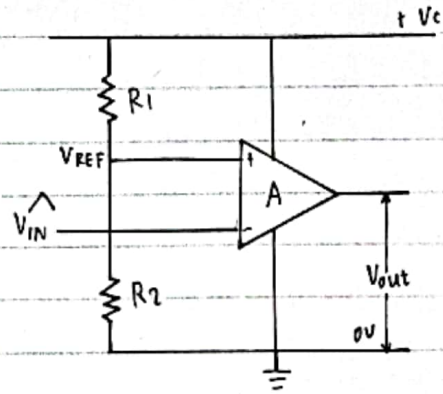
Rangkaian Non-Inverting Comparator
⟶ Positive Voltage Comparator



R_1 = 10 K Ohm

R_2 = 1 KOhm

Rangkaian Inverting Comparator
⟶ Negative Voltage Comparator



R_1 = 10 K Ohm

R_2 = 1 K Ohm

PERHITUNGAN TEORI

Rangkaian Non-Inverting Komparator

$$V_o = \begin{cases} V_{s+} & V_{in} > V_{ref} \\ V_{s-} & V_{in} < V_{ref} \end{cases}$$

Rangkaian Inverting Komparator

$$V_o = \begin{cases} V_{s-} & V_{in} < V_{ref} \\ V_{s-} & V_{in} > V_{ref} \end{cases}$$

$$V_{ref} = \frac{R_2}{R_1 + R_2} \cdot V_{cc}$$

$$= \frac{1K}{10K + 1K} \cdot 10\,V$$

$$= 0{,}9\,V$$

TABEL PENGUKURAN

Non-Inverting Komparator → Saat R_2 diganti potensio 10KΩ

Minimal → 2,6Ω $V_{REF} = 10\,mV$

Vref (V)		%
Teori	Praktek	Error
0,909	0,929	2,2

Inverting Komparator → Saat R_2 diganti potensio 10KΩ

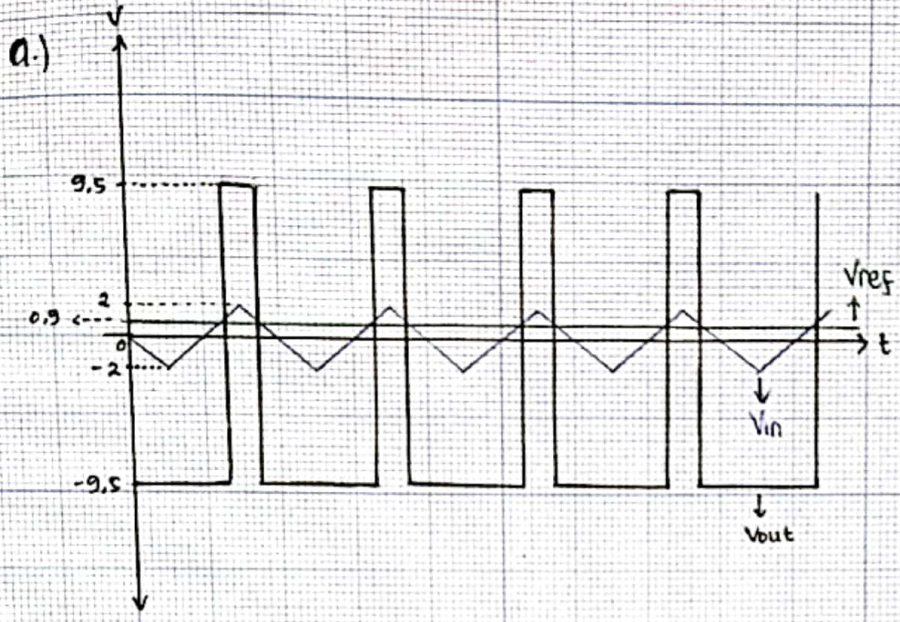
Minimal → 2,6Ω $V_{REF} = 13\,mV$

Vref (V)		%
Teori	Praktek	Error
0,909	0,929	2,2

✓

POSITIVE COMPARATOR

a.)



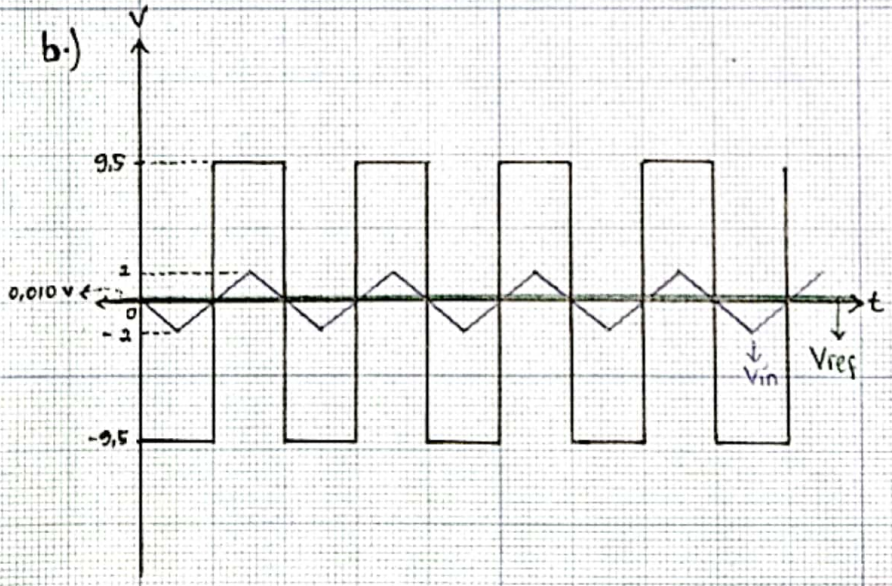
keterangan osc :
- volt/div : 5V/div
- time/div : 0,5 m/s

Diketahui :
- Vin = 4 Vpp
- F = 1 kHz

Hasil :
- Vout = 19 Vpp.
- Vref : 0,9 V

a.) Gambar sinyal Rangkaian Positive Comparator ketika $R_1 = 10k$, $R_2 = 1k\Omega$

b.)



keterangan osc :
- Volt/div : 5V/div
- time/div : 0,5 m/s

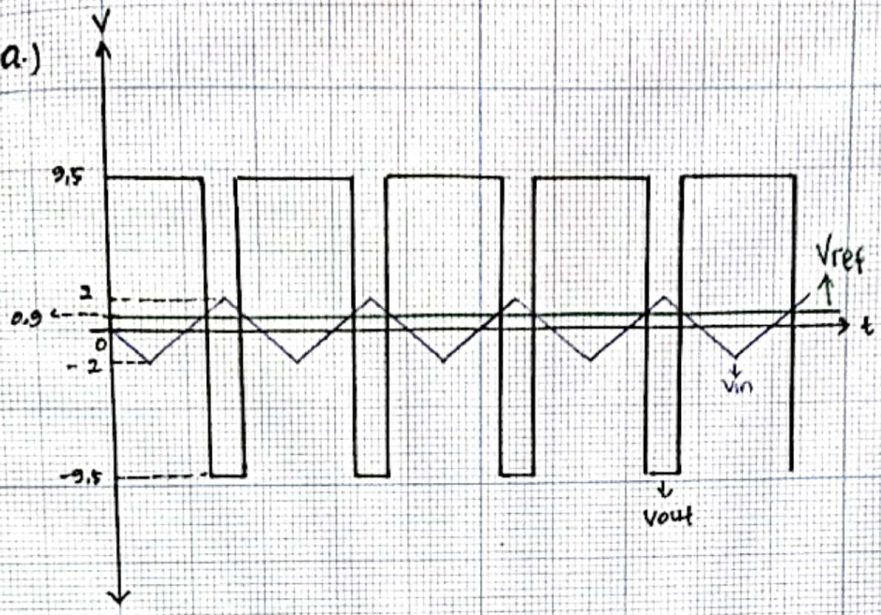
Diketahui :
- Vin = 4 Vpp
- F = 1 kHz

Hasil :
- Vout = 19 Vpp
- Vref = 10mV = 0,010V

b.) Gambar sinyal Rangkaian Positive Comparatore ketika menggunakan potensiometer yang diatur sebesar 2,6 Ω (minimal)

NEGATIVE COMPARATOR

a.)



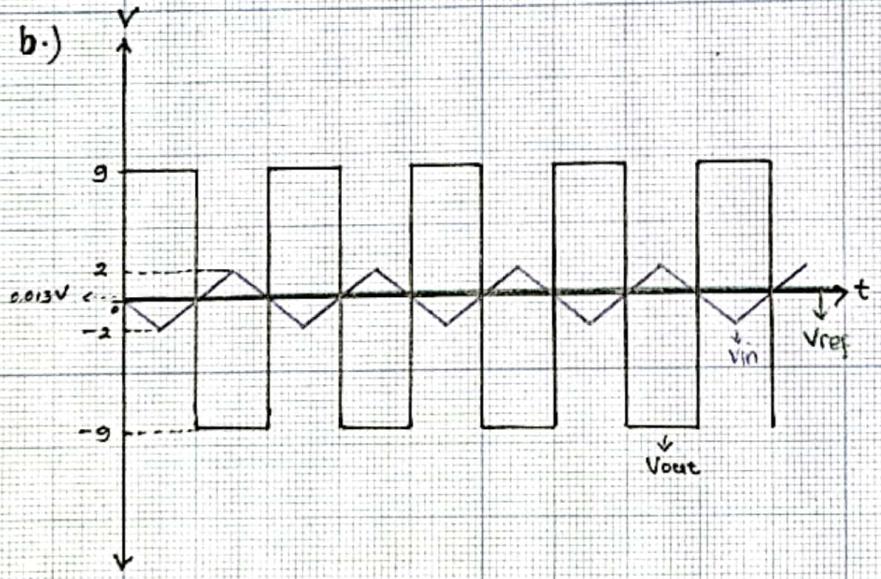
Keterangan osc :
- Volt / div = 5 V/div
- time / div = 0,5 m/s

Diketahui :
- Vin = 4Vpp
- F = 1KHz

Hasil :
- Vout = 19Vpp

a.) Gambar sinyal Rangkaian negative comparator ketika $R_1 = 10k\Omega$, $R_2 = 1k\Omega$

b.)



Keterangan osc :
- Volt / div = 5 V/div
- time / div = 0,5 m/s

Diketahui :
- Vin = 4Vpp
- F = 1KHz

Hasil :
- Vout = 18Vpp
- Vref = 13 mV
  = 0,013 V

b.) Gambar sinyal Rangkaian negative comparator ketika menggunakan potensiometer yang diatur sebesar 2,6 Ω (minimal)

JURNAL PRAKTIKUM ELEKTRONIKA ANALOG

Aplikasi Op-Amp Integrator dan Differentiator

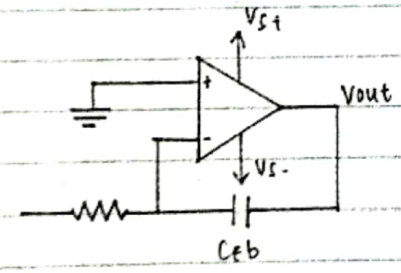
Tanggal Praktikum : Rabu, 23 November 2022
Kelompok : 3
Nama Anggota : 1. Anisha Fitri Pertiwi (21134400S)
2. Aulia Nadzif Pramesti (21134400G)
3. Bagea Alifa
4. Harun Maulana Ridho
5. Rezan Candrawinata
Kelas : 2NK

23/11

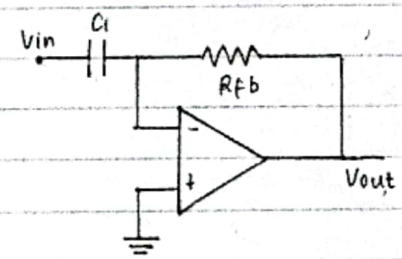
ALAT DAN BAHAN

1. DC Power Supply (2)
2. Function Generator (1)
3. Oscilloscope (1)
4. Op-Amp 741 (1)
5. R = 10 KOhm (1) ; R = 1 KOhm ; C = 0,1 µF
6. Multimeter (1)
7. Kabel Jumper (secukupnya)
8. Konektor Jumper (secukupnya)
9. Kertas HVS dan millimeter block (masing-masing 1 lembar / kelompok)

RANGKAIAN PERCOBAAN



Integrator



Differentiator

- Vin = 2 Vpp
- f = 1 KHz
- + Vcc = + 10 V
- - Vcc = - 10 V

PERHITUNGAN TEORI

1) Integrator

$$Vo = -\left(\frac{1}{R_1 C_{fb}}\right)\int_0^t Vin\, dt$$

input sinyal kotak (⊓⊔⊓)

$$Vo = -\left(\frac{1}{1000 \cdot 0{,}1 \cdot 10^{-6}}\right)\int_0^{0{,}0005} 1\, dt$$

$$= -10.000\ t\ ]_0^{0{,}0005}$$

saat $t = 0{,}0005 \rightarrow -5$ Vpp

saat $t = 0 \rightarrow 0$ Vpp

$t = 0 \rightarrow 0{,}0005$; $Vi = +1$ V

saat $t = 0{,}0005 \rightarrow +5$ Vpp

saat $t = 0 \rightarrow 0$ Vpp

$t = 0 \rightarrow 0{,}0005$; $Vi = -1$ V

input sinyal segitiga (∧∧∧)

$Vo = \frac{1}{2}$ Vo sinyal kotak

$= \frac{1}{2} \cdot 5V$

$= 2{,}5$ V ✓

2) Differentiator

$$Vo = -R_{fb} C_i \frac{dVin}{dt}$$

Input Sinyal Segitiga

$$= -1000 \cdot 0{,}1 \times 10^{-6} \cdot \frac{2V}{0{,}5 \cdot 10^{-3}}$$

$$= \frac{-2 \times 10^{-4}}{0{,}5 \times 10^{-3}}$$

$\Rightarrow -0{,}4$ V (untuk $t = 0 \rightarrow 0{,}0005$)

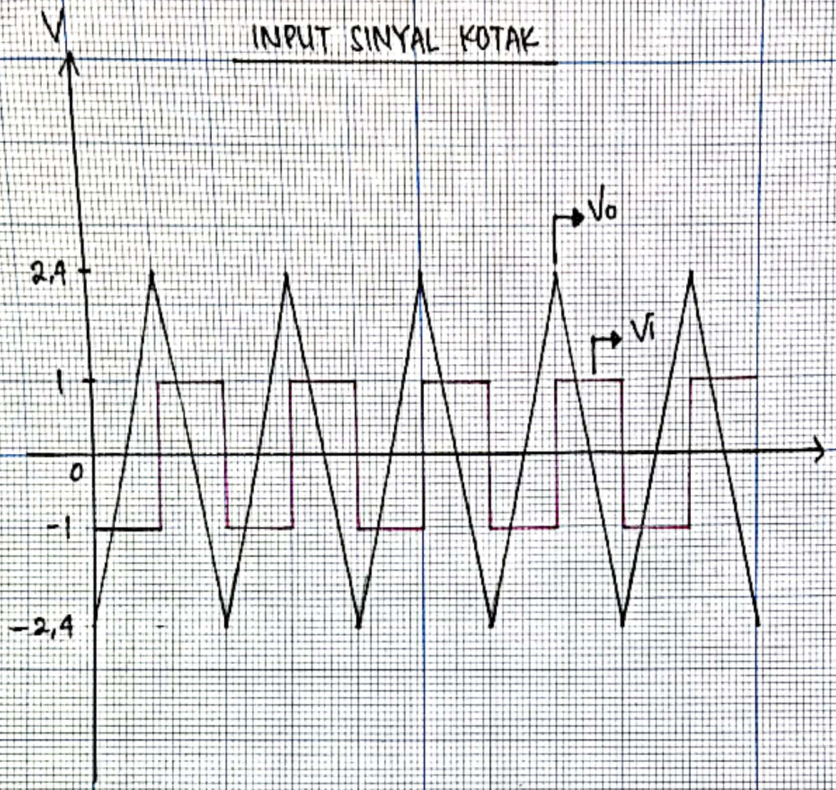
$\Rightarrow +0{,}4$ V (untuk $t = 0{,}0005 \rightarrow 0{,}001$)

TABEL PERCOBAAN

parameter	Intergrator			differentiator		
	teori	praktek	error	teori	praktek	error
	Input kotak			Input kotak		
V_i	2Vpp	2Vpp	0%	2Vpp	2Vpp	0%
V_o	5Vpp	4,8Vpp	4%		14 Vpp	
	Input Segitiga			Input Segitiga		
V_i	2Vpp	2Vpp	0%	2Vpp	2Vpp	0%
V_o	2,5Vpp	2,4Vpp	4%	0,8Vpp	0,8Vpp	0%

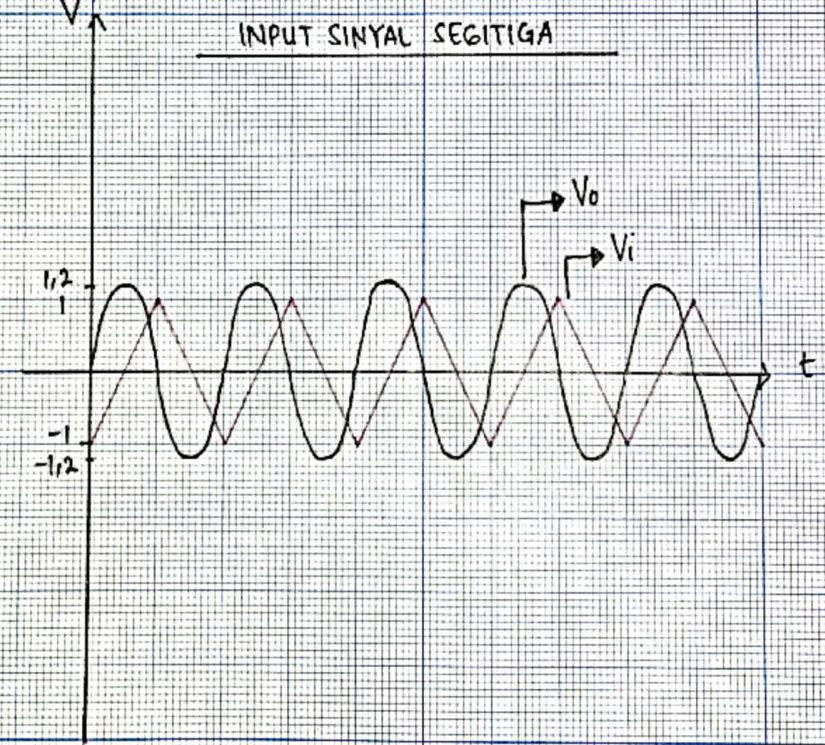
INTEGRATOR

INPUT SINYAL KOTAK



Ket:
Volt/div = 1V
Time/div = 0,5ms

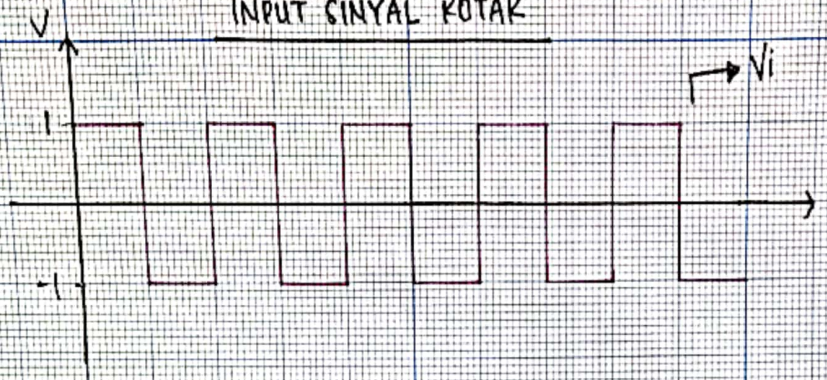
INPUT SINYAL SEGITIGA



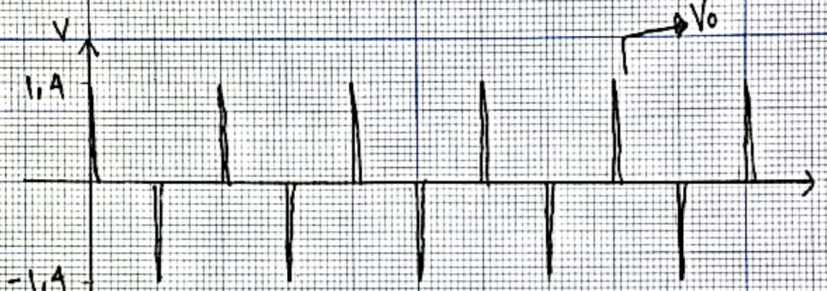
Ket:
Volt/div = 1V
Time/div = 0,5ms

DIFFERENTIATOR

INPUT SINYAL KOTAK

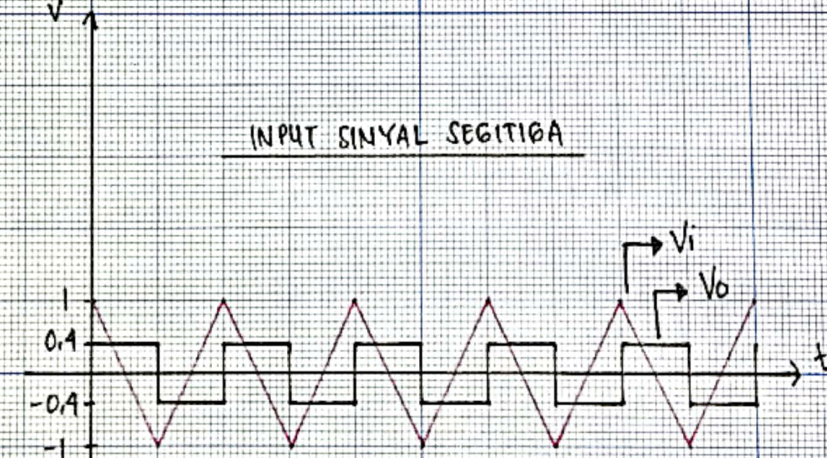


Ket :
Volt/div = 1V
Time/div = 0,5ms



Ket :
Volt/div = 5V
Time/div = 0,5ms

INPUT SINYAL SEGITIGA



Ket :
Volt/div = 1V
Time/div = 0,5ms

MODUL PRAKTIKUM ELEKTRONIKA ANALOG
Oscillator

Tanggal Praktikum : Rabu, 30 November 2022

Kelompok : 3

Nama : 1. Anisha Fitri Pertiwi (21134400S)
2. Aulia Nadzif Pramesti (21134400G)
3. Bagea Alifa
4. Bahrani
5. Rezan Candrawinata

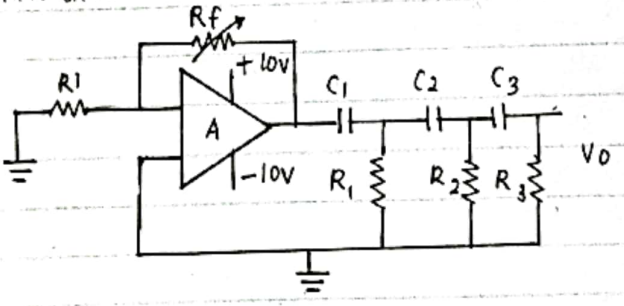
Kelas : 2NK

30/11

ALAT DAN BAHAN

1. DC Power Supply : 2 buah
2. Milimeter block : 1 buah
3. Oscilloscope : 1 buah
4. IC u A741 : 1 buah
5. Resistor 1kΩ : 4 buah
6. Resistance Decade Box : 1 buah
7. Copasitor 0,01 MF : 3 buah
8. Multimeter : 2 buah
9. Kabel Jumper : secukupnya
10. Konektor Jumper : secukupnya
11. Kertas HVS : secukupnya

RANGKAIAN PERCOBAAN



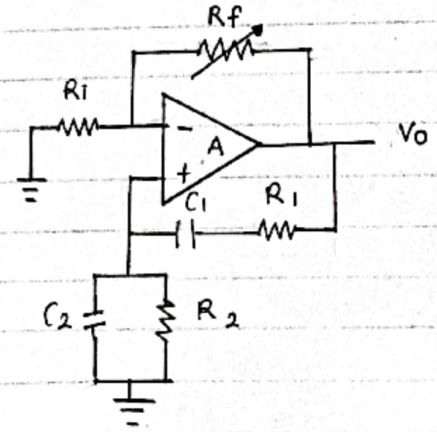
Rangkaian Phase Shift Oscillator

$C_1 = C_2 = C_3 = C = 0{,}01\ \mu F$

$R_1 = R_2 = R_3 = R = 1\ k\Omega$

$R_i = 1\ k\Omega$

R_f = < Potensio 100K Ω)



Rangkaian Wien Bridge Oscillator

$C_1 = C_2 = C = 0{,}01\ \mu F$

$R_1 = R_2 = R = 1k\Omega$

$R_i = 1\ k\Omega$

R_f = (Potensio 10K Ω)

ONGAN TEORI

hase Shift Oscillator

$A = 29 \times$ $\qquad R_f = R_1 \times A = 10^3 \times 29 = 29K\Omega$

$B = \frac{1}{29}$

$$f_o = \frac{1}{2\pi RC\sqrt{6}}$$

$$= \frac{1}{2\pi \cdot 1 \cdot 10^3 \cdot 0{,}01 \cdot 10^{-6} \cdot \sqrt{6}}$$

$$= 6{,}497\ KHz$$

o) Wien Bridge Oscillator

$A = 2x$ $\qquad R_f = R_1 \times A = 10^3 \times 2 = 2K\Omega$

$B = \frac{1}{3}$

$$f_o = \frac{1}{2\pi RC}$$

$$= \frac{1}{2\pi \cdot 1 \cdot 10^3 \cdot 0{,}01 \cdot 10^{-6}}$$

$$= 15{,}915\ KHz$$

TABEL PERCOBAAN

	Phase shift osc		wien bridge osc	
	fo	Rf	fo	Rf
Teori	6,497 KHz	29KΩ	15,915KHz	2KΩ
praktek	5KHz	78,7KΩ	13,157KHz	2kΩ
error	23%	167%	17%	0%

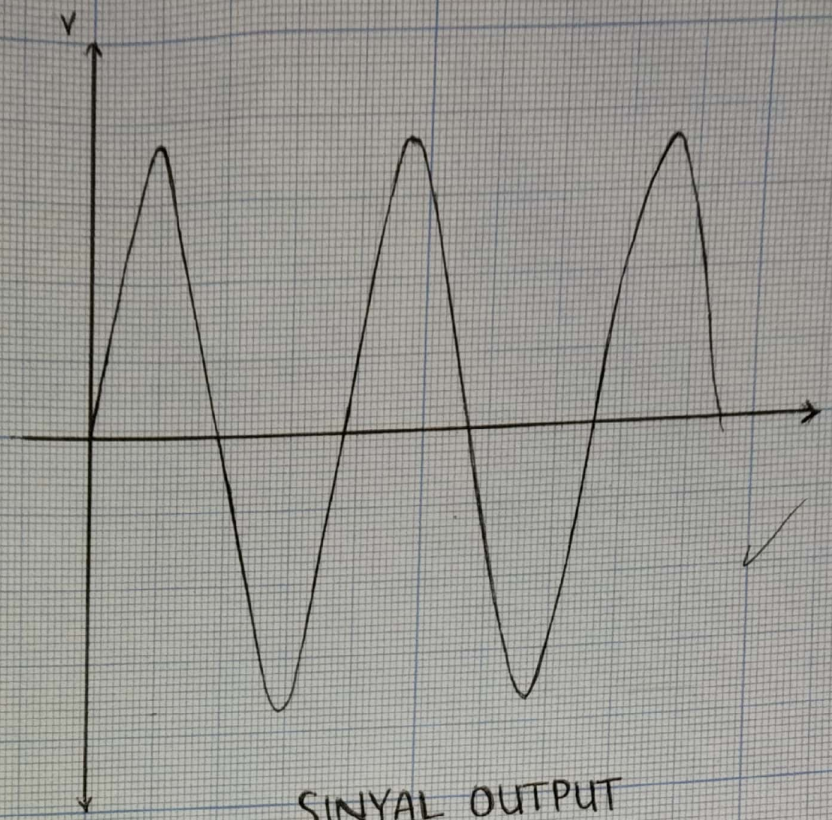
Wein Bridge Osc. < R potensio 10KΩ >

$$f_o = \frac{1}{T} = \frac{1}{\text{div. vertikal} \times t/div} = \frac{1}{3{,}8 \times 20.10^{-6}} = 13{,}157\ KHz$$

Phase Shift Osc. < R potensio 100 KΩ >

$$f_o = \frac{1}{T} = \frac{1}{\text{div. vertikal} \times t/div} = \frac{1}{4 \times 50.10^{-6}} = 5\ KHz$$

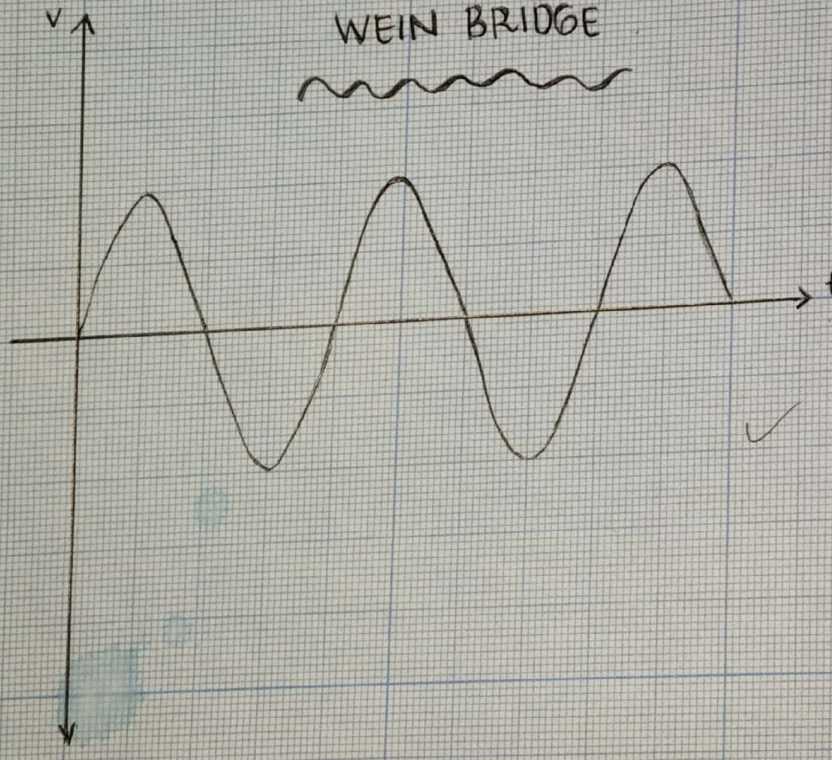
SINYAL OUTPUT PHASE SHIFT



keterangan :

- Volt/div = 2V
- time/div = 20μs

SINYAL OUTPUT WEIN BRIDGE



keterangan :

- Volt/div = 20mV
- time/div = 20μs